MANUEL PEREZ GIL

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Lubbock, TX

EDUCATION & CERTIFICATIONS

Texas Tech University Lubbock, TX
Bachelor of Science, Computer Science | GPA: 3.51 *May 2027*

- Coursework: Generative AI, Software Engineering, Autonomous Driving, Data Structures & Algorithms, OOP
- **CodePath AI301: Open Source Capstone** – In Progress, Fall 2026
- **AI 201: Applications of AI Engineering** – CodePath, Aug 2026
- **AWS Cloud Quest:** Cloud Practitioner Training Badge — AWS, Jun 2026

PROJECTS

Pipeline Sentinel | *Java, Spring Boot, Next.js, PostgreSQL, Docker, AWS Lightsail*

- Built a full-stack CSV validation platform with a TypeScript/React (Next.js) dashboard and Java/Spring Boot REST APIs to inspect validation results and quarantined records
- Persisted validation runs, row-level issues, and quarantined records in PostgreSQL using JPA and Flyway to maintain an audit trail of transaction batches
- Deployed to AWS Lightsail using Docker Compose and Caddy HTTPS; configured GitHub Actions CI for 20 backend tests, 8 frontend tests, package, lint, and build checks

LiveQuiz | *Next.js, TypeScript, DynamoDB, AWS Lambda, DynamoDB Streams, API Gateway WebSocket, Vercel*

- Built a real-time multiplayer trivia platform with Next.js, TypeScript, and a single-table DynamoDB design for 4 entity types: rooms, players, questions, and answers
- Synchronized concurrent game sessions using DynamoDB Streams, AWS Lambda, and API Gateway WebSocket, achieving sub-500 ms state propagation without persistent servers
- Integrated the Claude API to generate trivia on demand, parse structured JSON responses, and store game-ready questions in DynamoDB at game start; hosted the frontend on Vercel

IncidentPilot | *TypeScript, Node.js, Slack Bolt, MCP SDK*

- Built a TypeScript/Node.js Slack incident agent using Slack Bolt to classify production alerts into 4 severity tiers across 6 services with a rule-based pipeline
- Implemented an MCP SDK server and client with 3 tools that used the Slack Web API to create incident channels, post structured briefings, and page the on-call engineer

EXPERIENCE

IT Support Technician, IT Help Central,TX Aug 2025 – Present

- Managed 20+ weekly ServiceNow tickets across 35+ Windows and macOS classrooms, maintaining 98% uptime through hardware troubleshooting, machine reimaging, and software configuration

Research Assistant, Interactive Data Visualization Lab (IDVL) May 2025 – Jan 2026

- Built a Python/Dash and Plotly analytics application for the 2025 IEEE SciVis Contest to explore 324,632 aluminum alloy compositions across 70 material properties
- Converted a 182 MB TSV dataset into 93 MB Parquet through a preprocessing pipeline, reducing storage by 48.7% and improving data access for interactive analysis
- Applied PCA to reduce 70 features to 2 interpretable components across 15 interactive panels with linked biplots and parallel-coordinate visualizations

ADDITIONAL

Languages: Spanish (Native), English (Fluent), Portuguese (Intermediate)
Leadership & Involvement: SHPE @ TTU – Member; 2026 National Convention Delegate | Texas Tech Soccer Club – Member, 2026–Present